

HEC-RAS 2D Dynamic River Simulation Guide

Technical Documentation

May 29, 2026

1 Introduction

This document outlines the workflow for conducting a 2D unsteady flow (dynamic) simulation in HEC-RAS using a topobathymetric GeoTIFF and known water surface elevations (heads).

2 Part 1: Setting Up the Terrain

Before performing hydraulic calculations, HEC-RAS requires a Digital Elevation Model (DEM) in a specific format (.hdf).

2.1 Open HEC-RAS:

Create a new project.

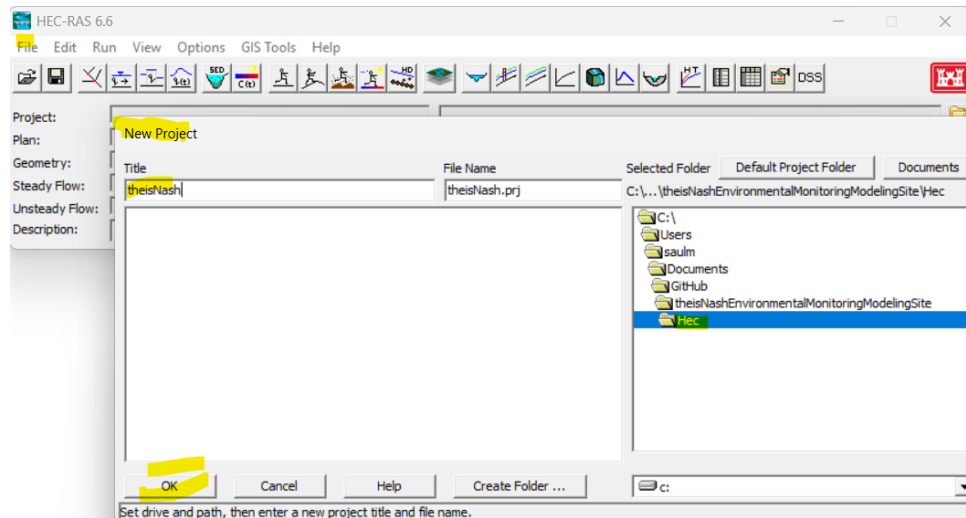


Figure 1: Create project

But be careful of the unit system

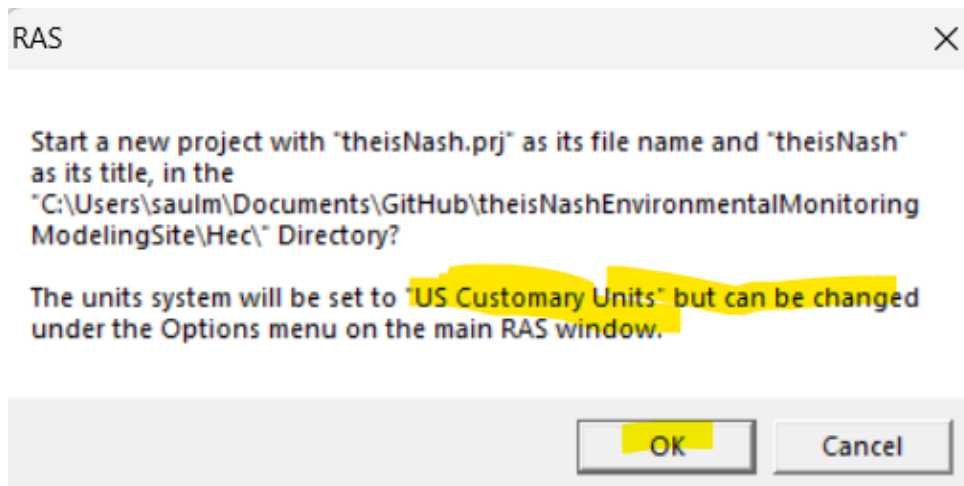


Figure 2: Warning over the unit system

Before importing shapefile its time to change the unit system of the project. Go to Options > Unit System (US Customary or System International).Select System International (SI): Meters, kilometers, etc.

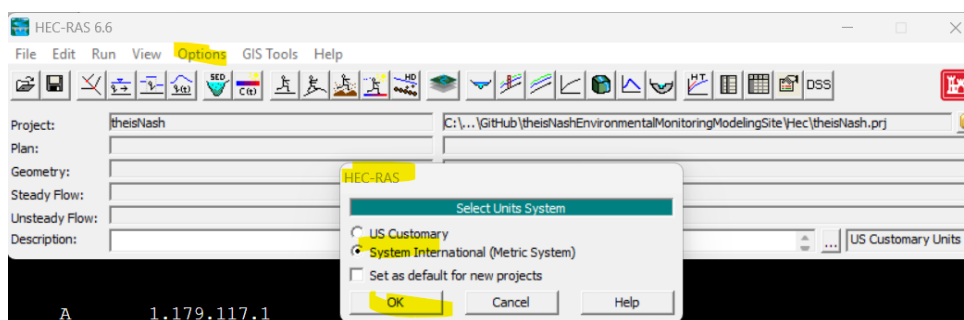


Figure 3: Select Units System

2.2 RAS Mapper:

Open the RAS Mapper tool.

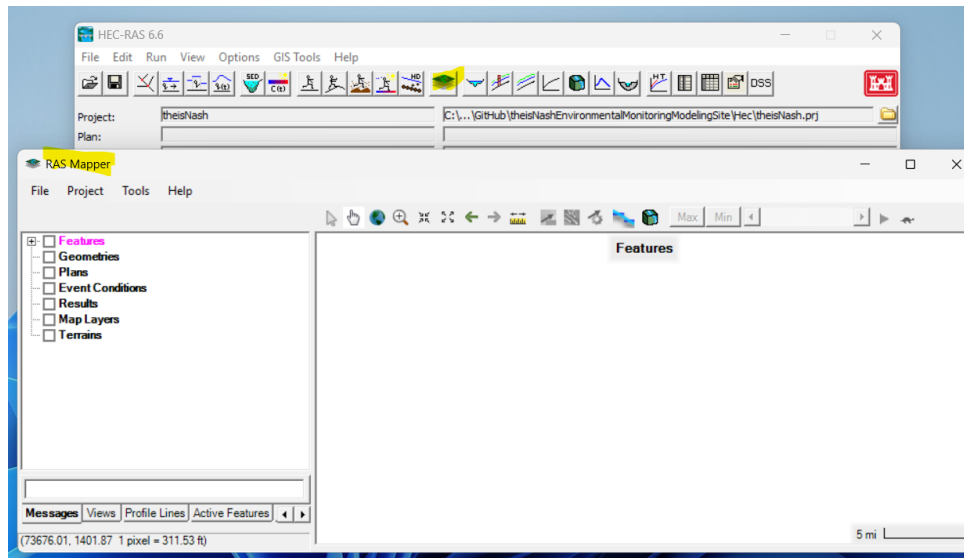


Figure 4: Ras Mapper tool

2.2.1 Coordinate System:

Go to *Project > Set Projection*. Select a Projection file (.prj) that matches your GeoTIFF's coordinate system.

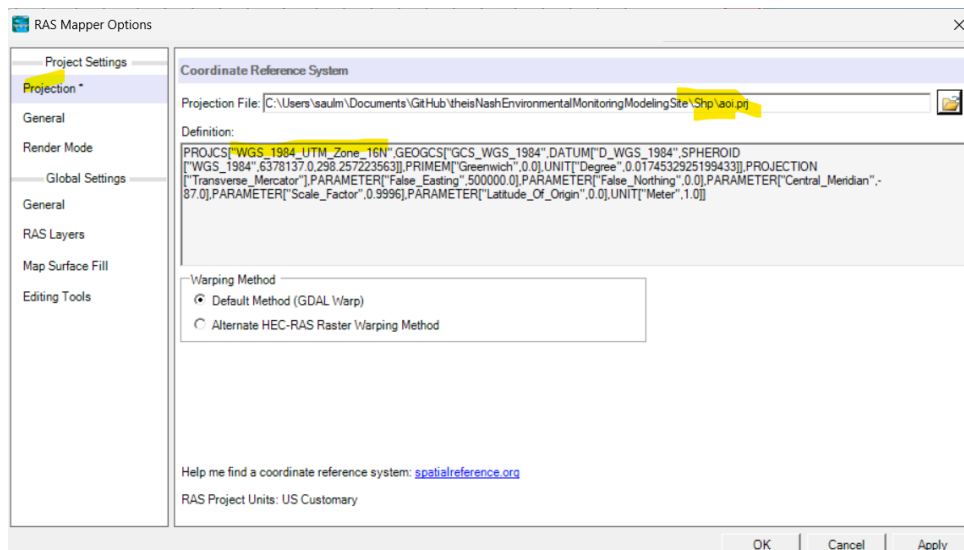


Figure 5: Set projection

2.2.2 Create Terrain:

Right-click on *Terrains* and select *Create New RAS Terrain*.

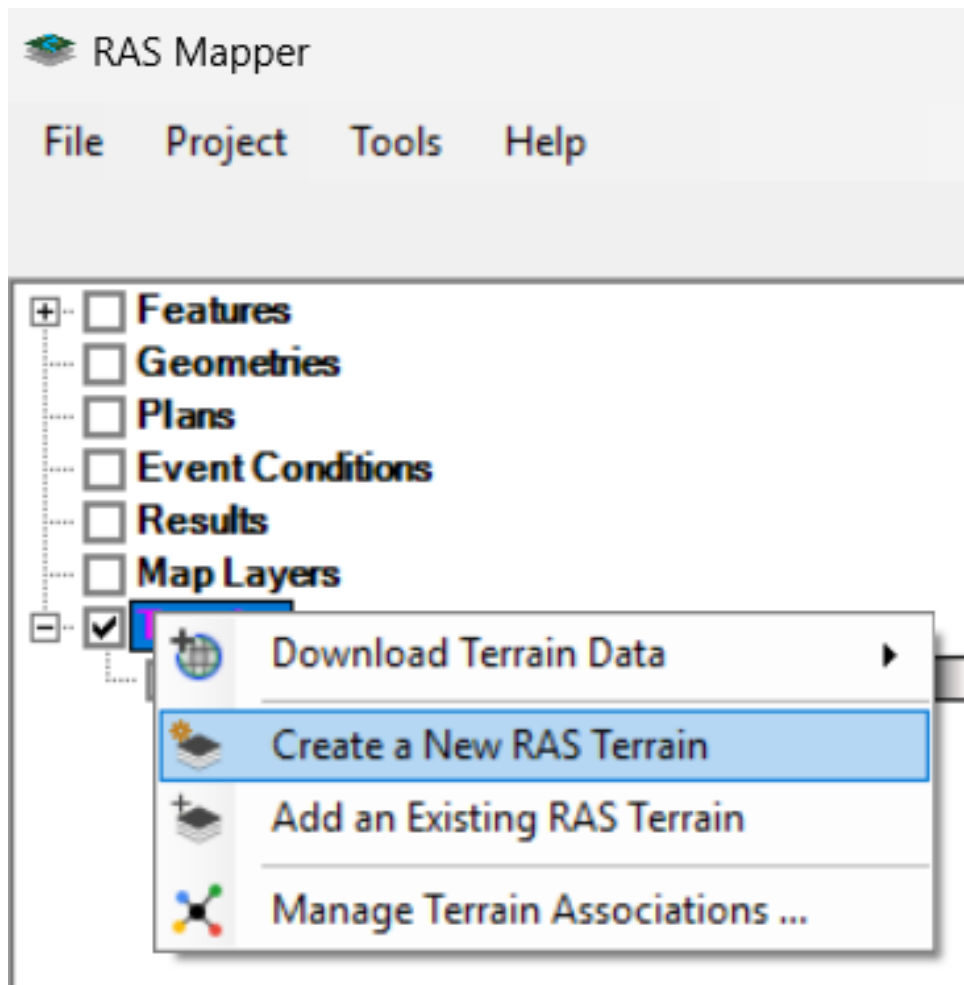


Figure 6: Create New RAS Terrain

Add your topobathymetric .tif file. This will convert the GeoTIFF into a HEC-RAS Terrain layer.

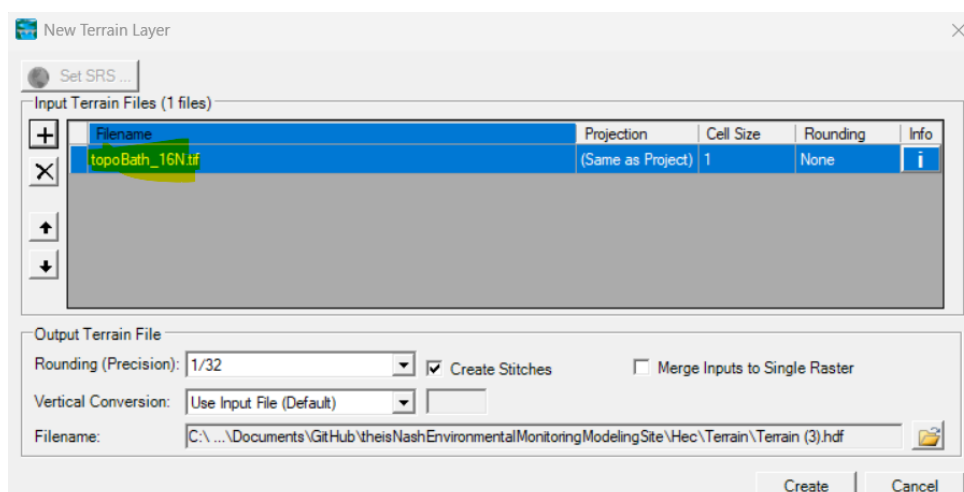


Figure 7: Add your tif file

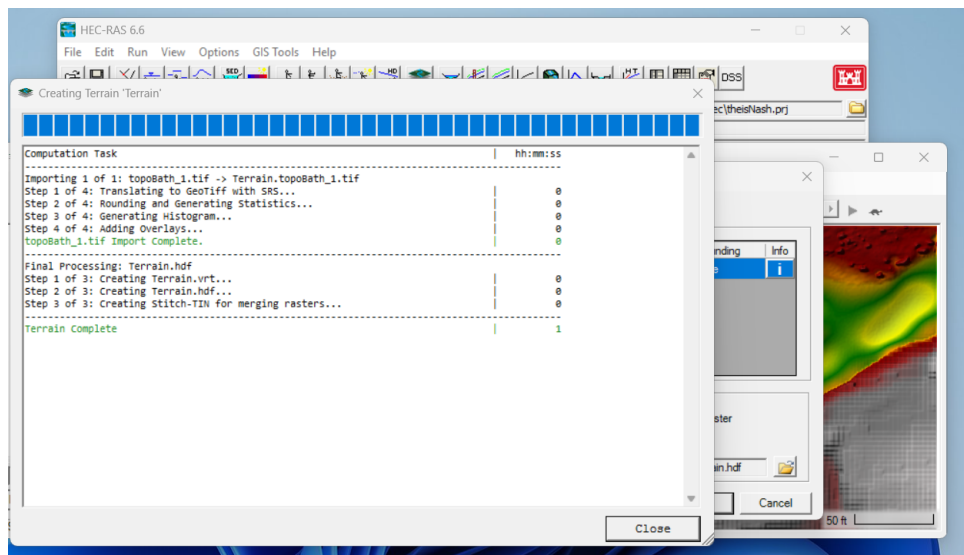


Figure 8: Hdf creation

3 Part 2: Defining the 2D Flow Area

3.0.1 Create Geometry:

In RAS Mapper, right-click *Geometries* and select *Add New Geometry*.

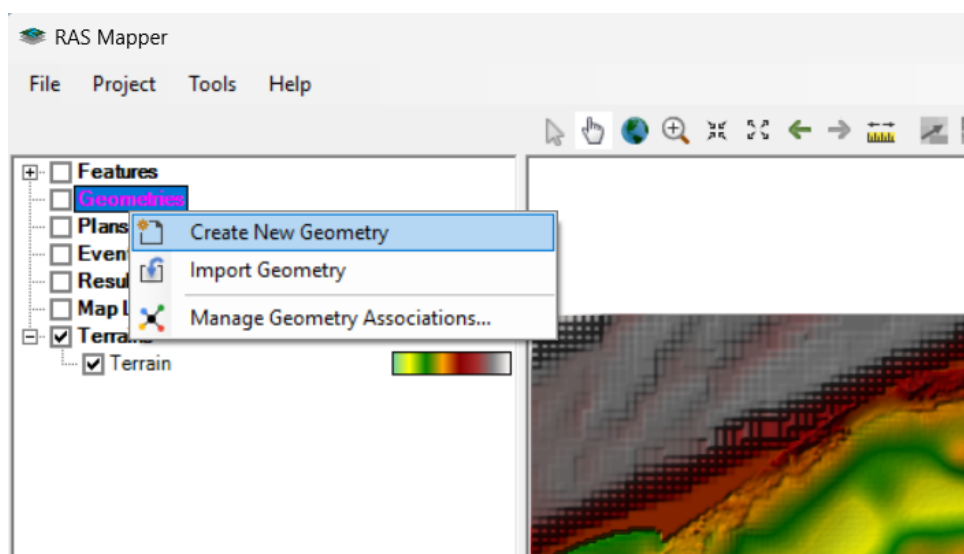


Figure 9: Add New Geometry

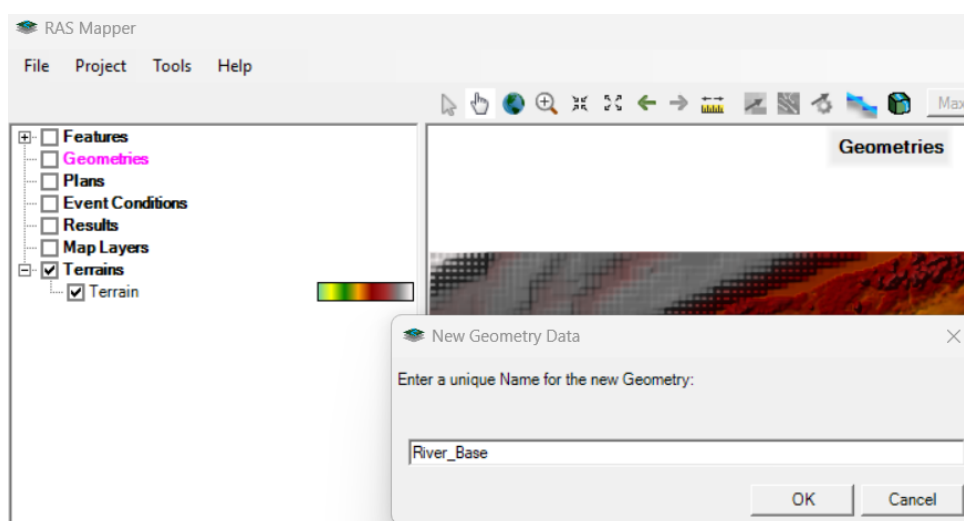


Figure 10: New Geometry Data

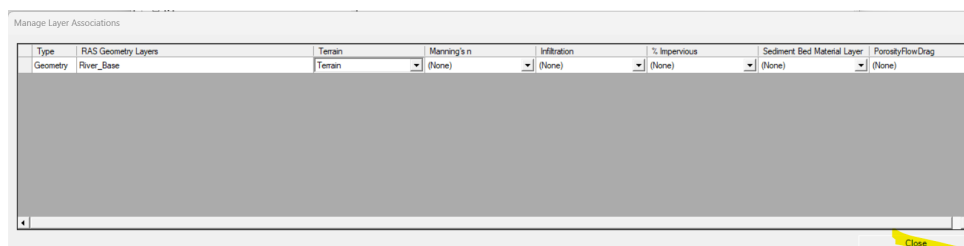


Figure 11: Manage Layer Associations

3.0.2 Import the Shapefile into a Layer

You cannot simply "open" a shapefile as a geometry; you must import its features into the specific HEC-RAS layer (like the 2D Flow Area / Perimeter).

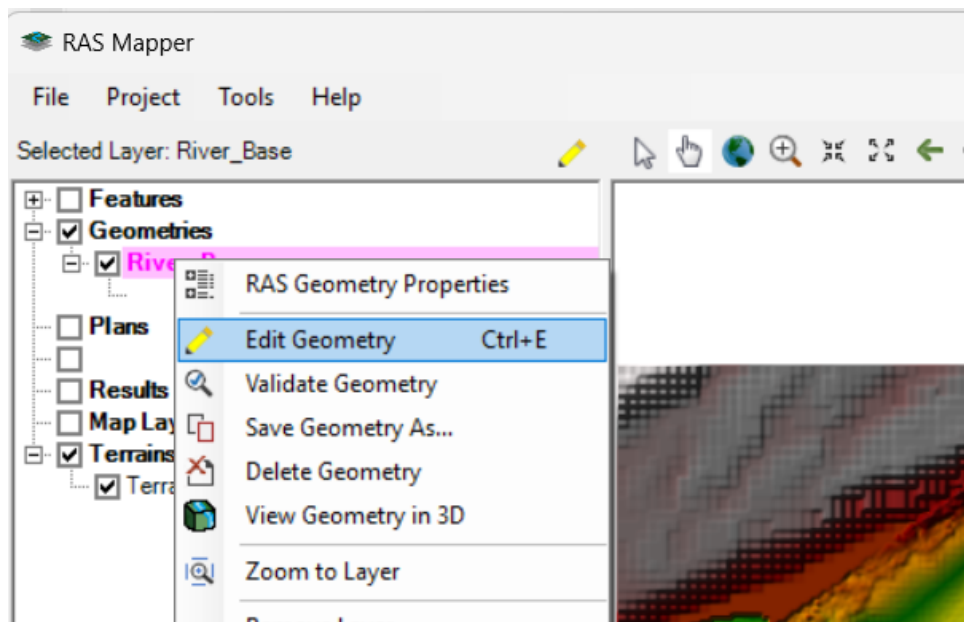


Figure 12: Edit geometry

3.0.3 Create Geometry:

Right-click on the specific sub-layer you want to populate (e.g., 2D Flow Areas / Perimeters).

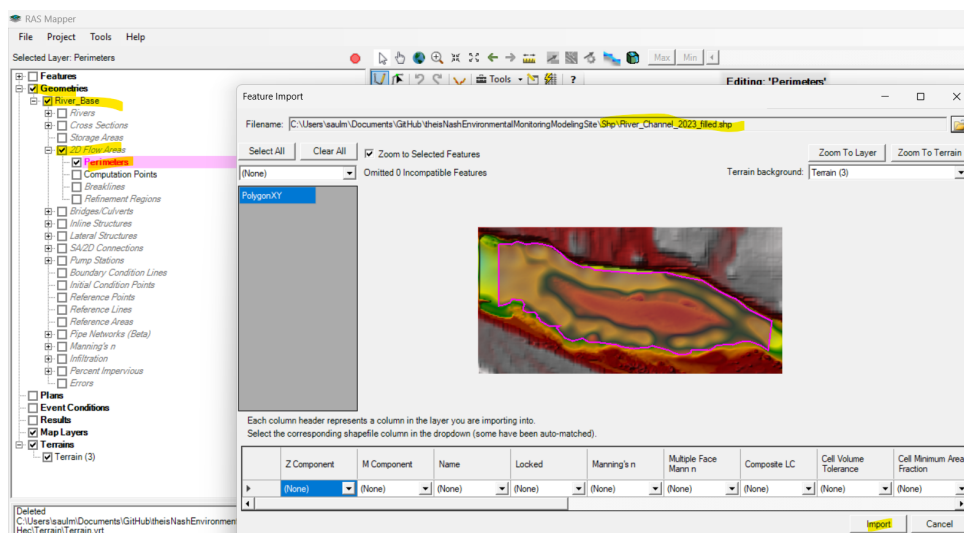


Figure 13: Feature Import

Right-click the 2D area polygon and select *Edit 2D Flow Area Properties*.

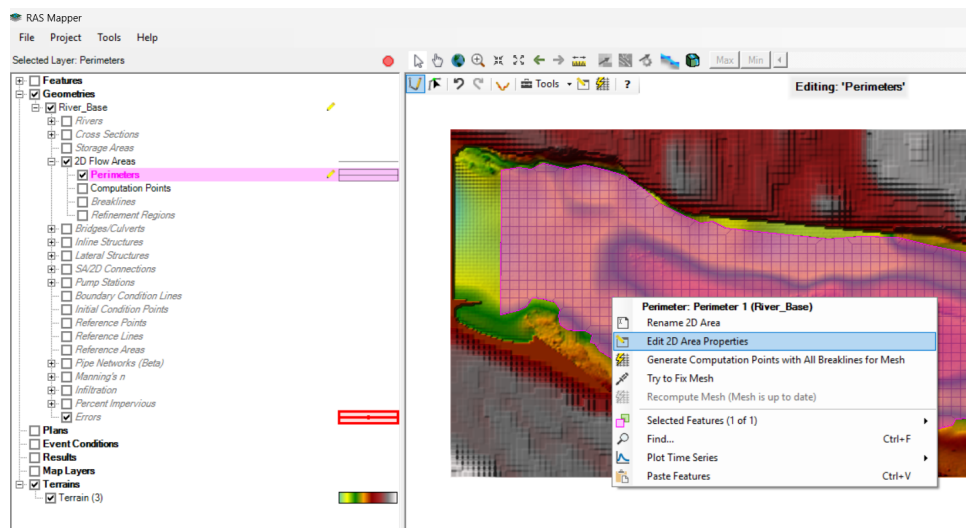


Figure 14: Edit 2D Flow Area Properties

Define your cell size (e.g., 2m x 2m). Click *Generate Computation Points*.

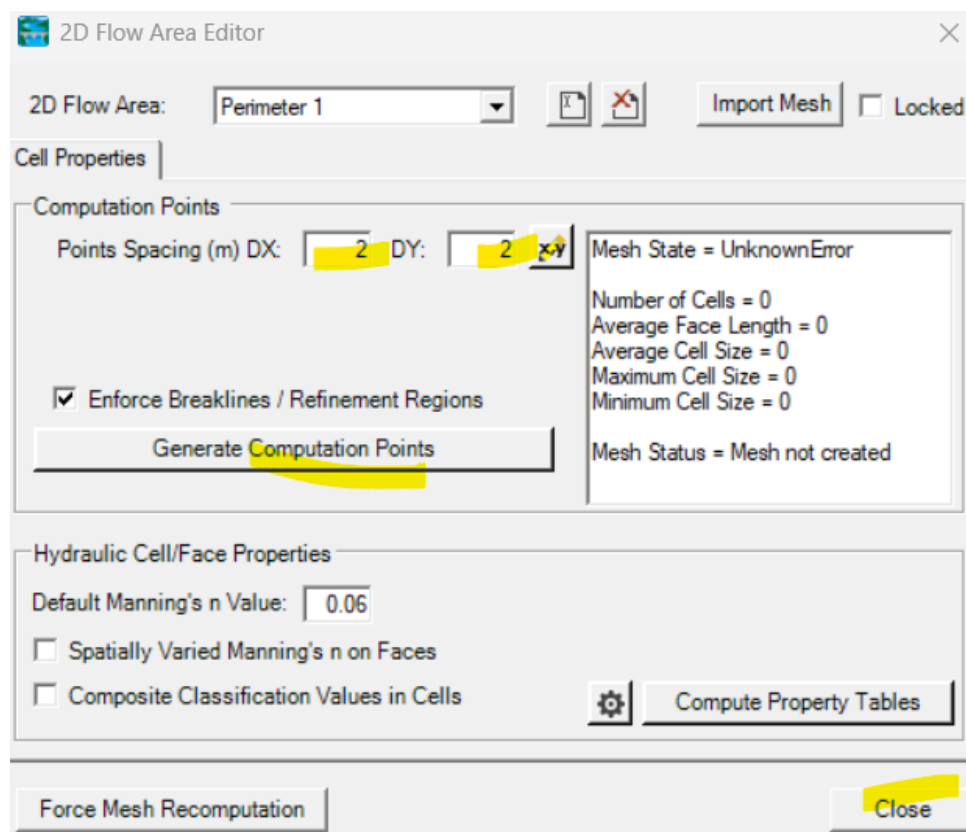


Figure 15: Edit 2D Flow Area Properties

4 Part 3: Boundary Conditions

Since you have "heads" (Water Surface Elevations - WSE), you will use these as boundary conditions.

4.1 Boundary Condition Lines

In RAS Mapper, select the *Boundary Condition Lines* layer under your Geometry.

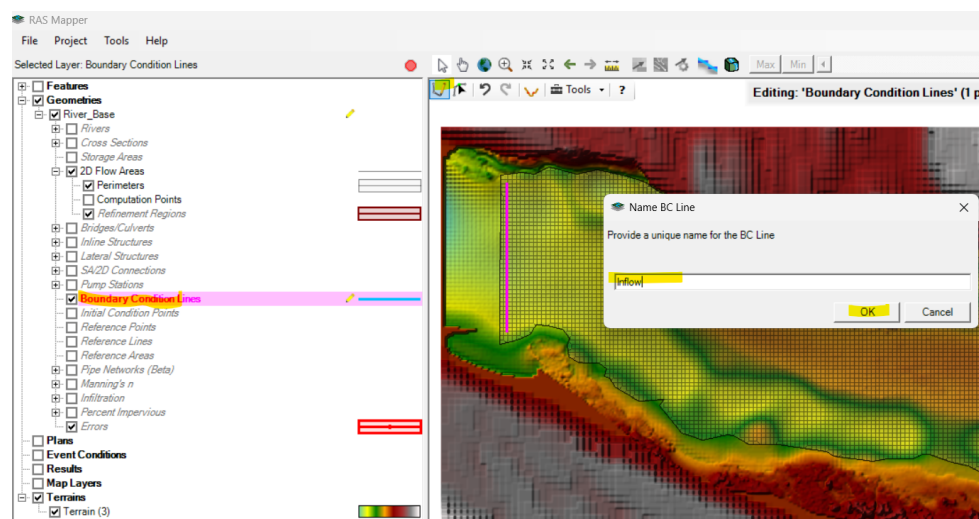


Figure 16: Name BC Line - Inflow

The outflow boundary condition has to be outside the model on the downstream direction.

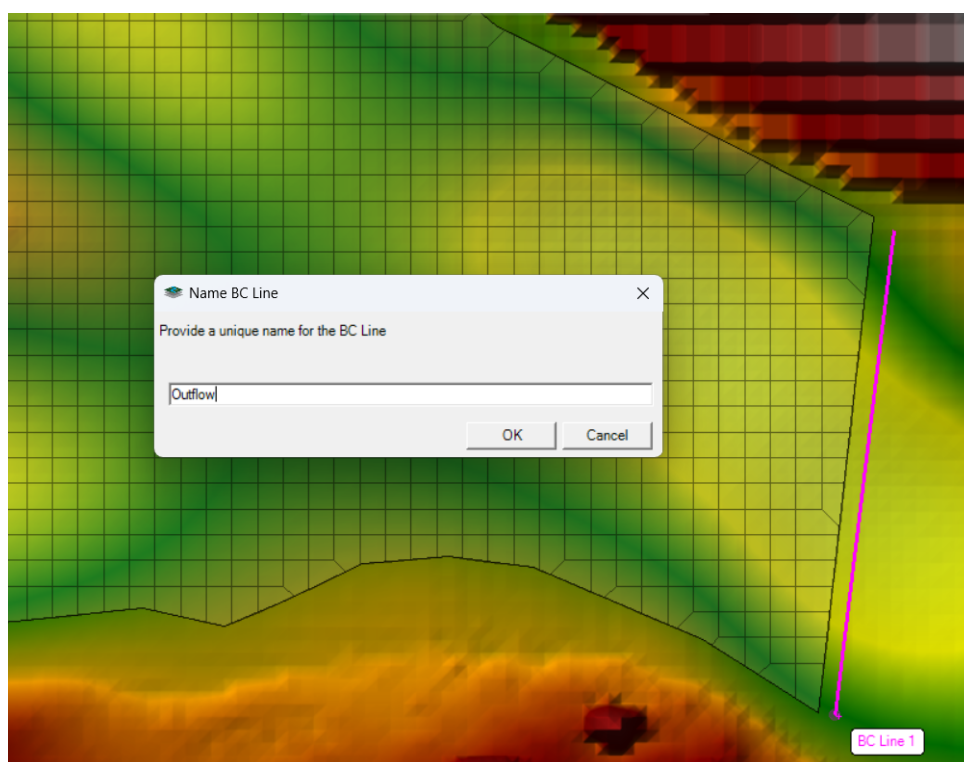


Figure 17: Name BC Line - Outflow

Close the editing mode for the recomputation of the line connectivity

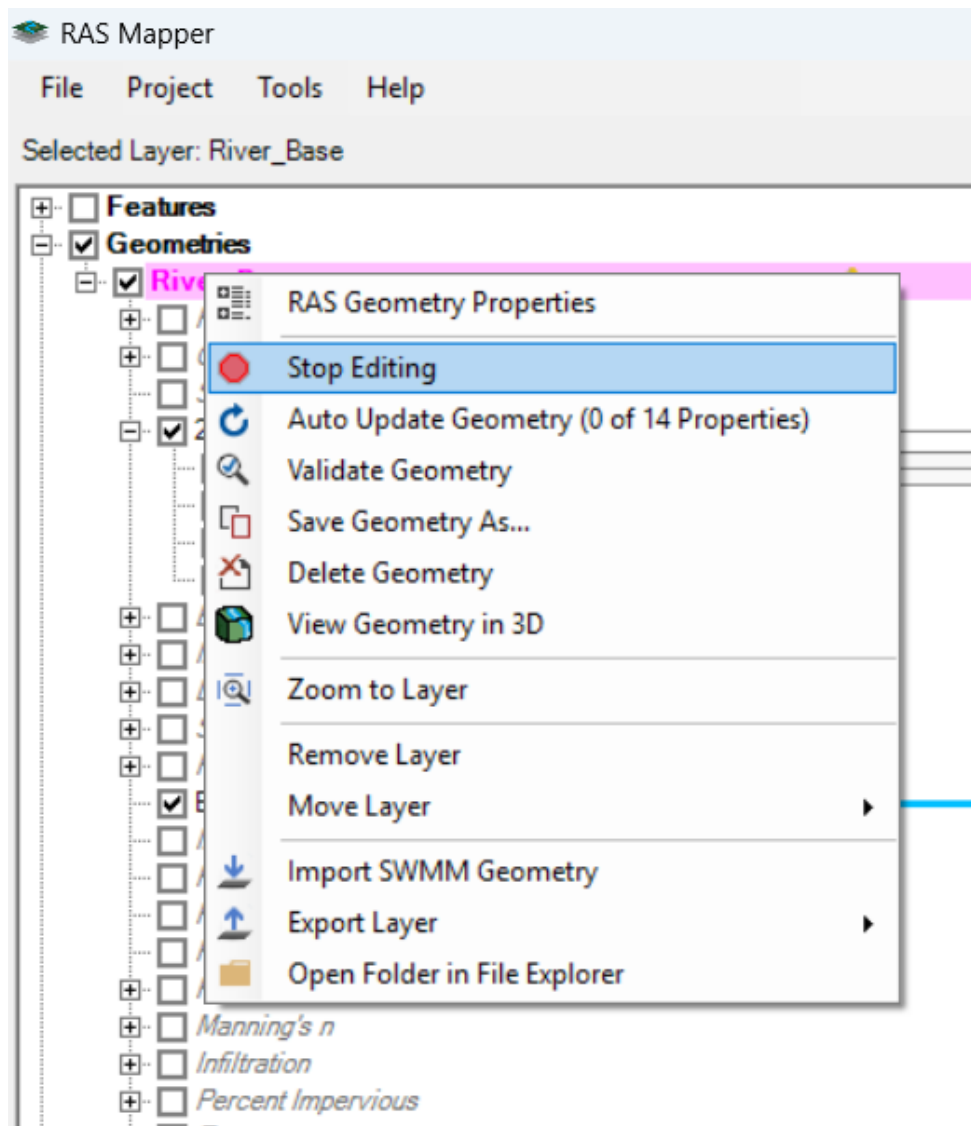


Figure 18: Close editing mode

Open attribute table of the boundary condition lines. Check that Inflow is set as Internal and Outflow is set as External.

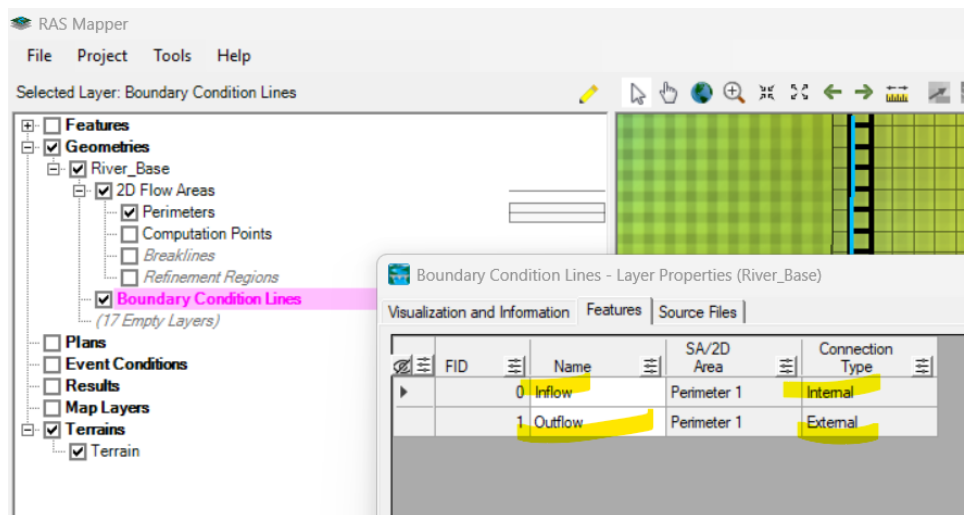


Figure 19: Layer Properties of Boundary Conditions

You can save your RAS Mapper project just in case.

5 Set up geometry in the HEC RAS project

In HecRas open Edit / Geometric Data

Open the saved geometry. The geometry should appear in the Geometric Data window and the path should appear on the HEC RAS project.

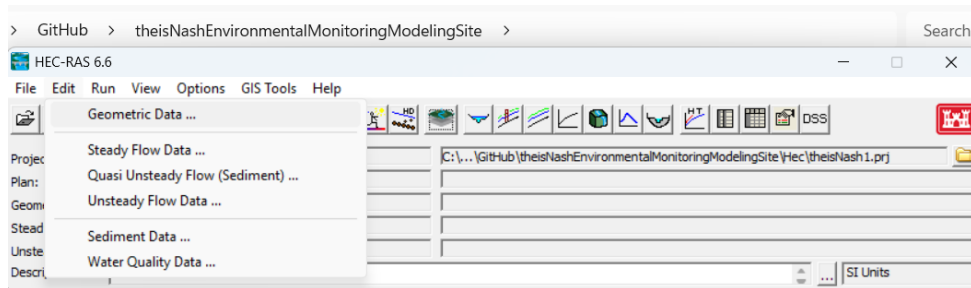


Figure 20: Edit Geometric Data

We open the saved geometric data to check that is ok and also to ensure that the HEC project recognizes this geometry.

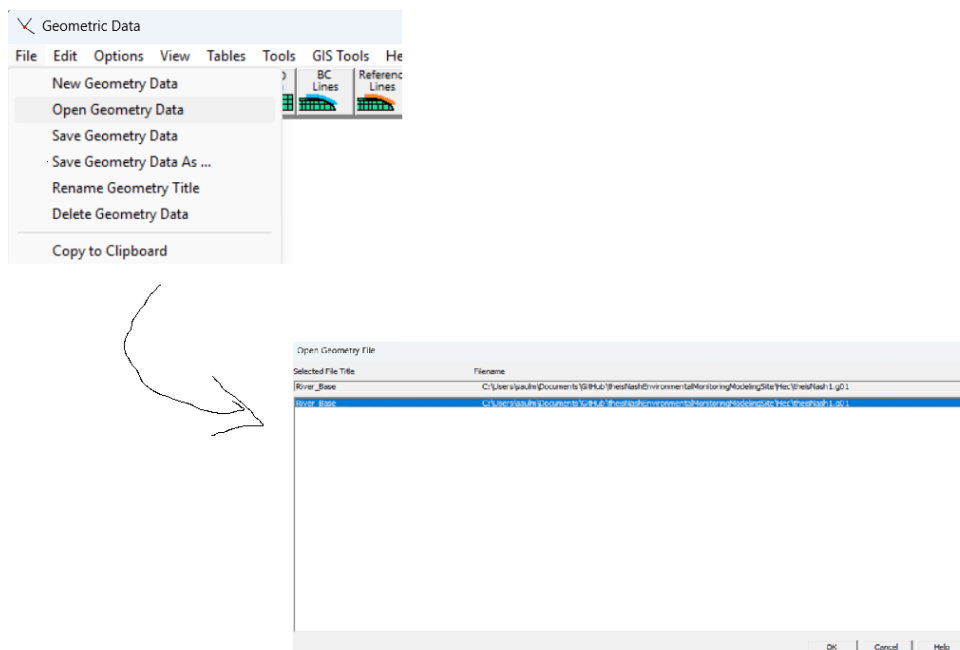


Figure 21: Open the saved geometry

Imported geometry in the HEC project.

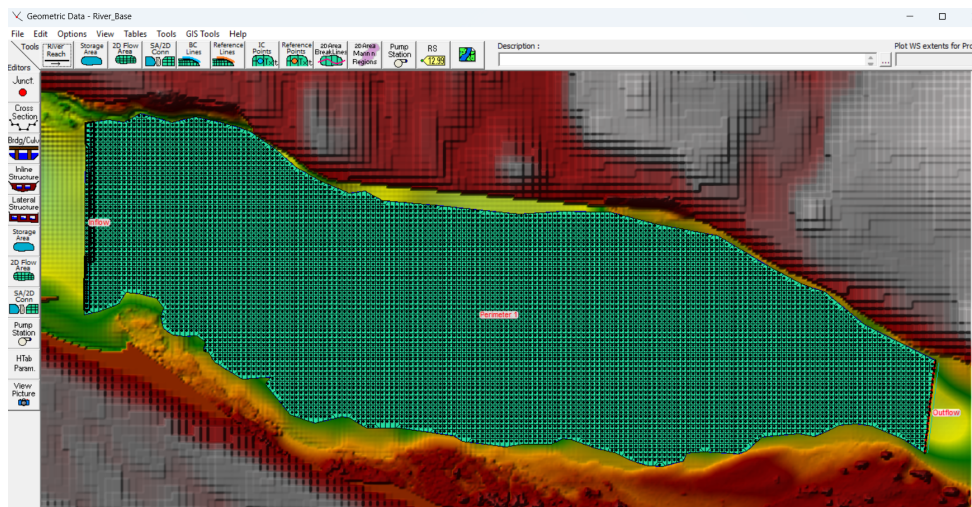


Figure 22: Imported geometry in the HEC project

Close the Geometric Data window and you will see that the RiverBase geometry is part of the HEC RAS project.

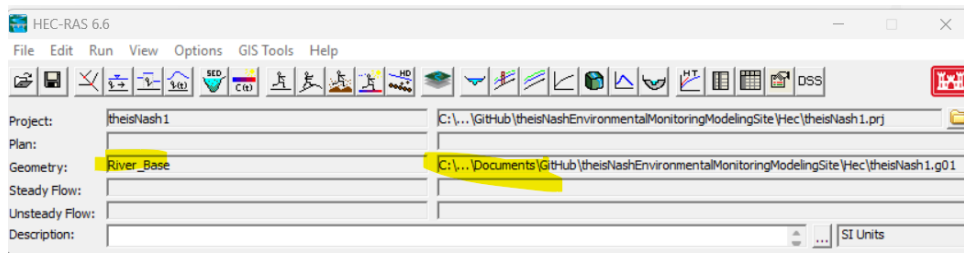


Figure 23: HEC RAS project with the geometry

5.1 Define Unsteady Flow Data

Press the view/edit unsteady flow data bottom. You will see the following.

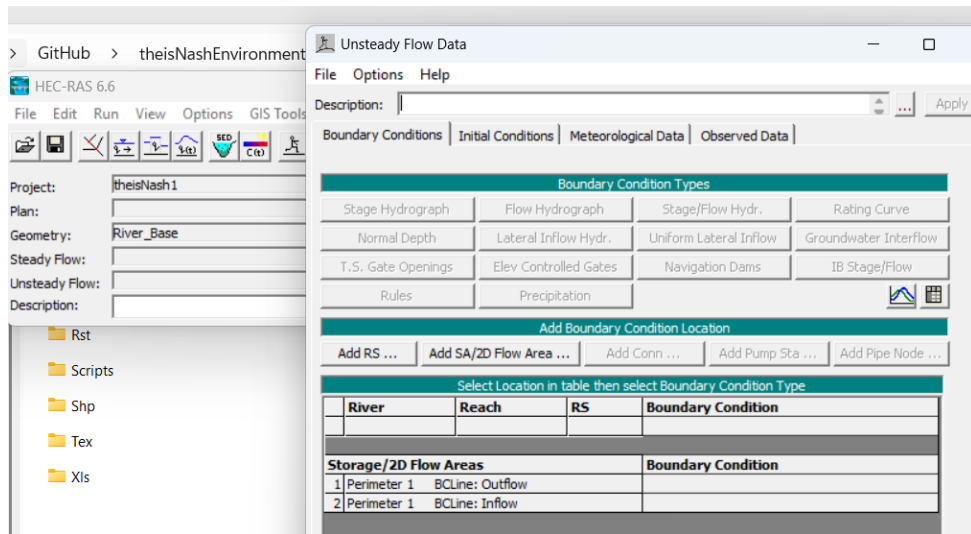


Figure 24: Unsteady Flow Data

For the Inflow boundary condition we are going to set up a flow hydrograph. Flow data is imported from a DSS file. Check the plot to see if imported correctly.

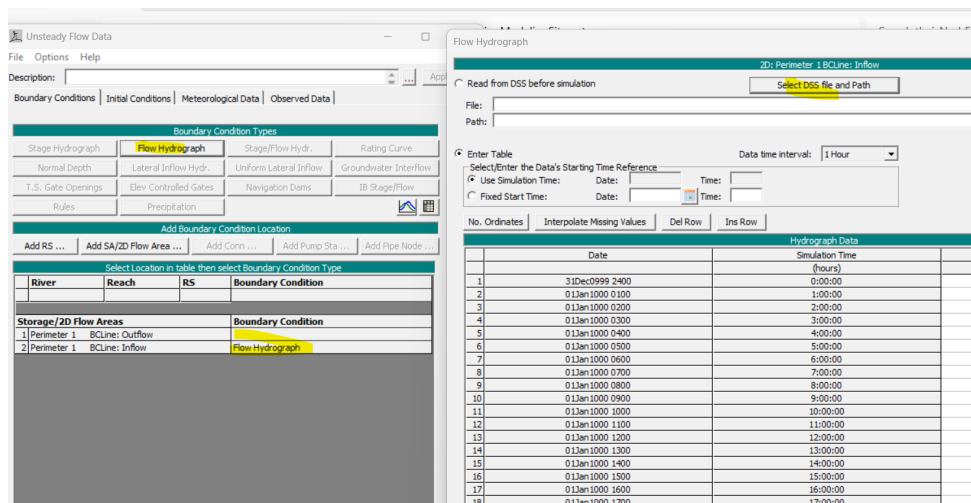


Figure 25: Flow of Inflow Boundary Condition



The screenshot displays the 'Unsteady Flow Data' dialog box in HEC-RAS. The 'Boundary Condition Types' tab is active, showing a grid of options where 'Normal Depth' is highlighted. Below this, the 'Add Boundary Condition Location' section contains buttons for 'Add RS ...', 'Add SA/2D Flow Area ...', 'Add Conn ...', 'Add Pump Sta ...', and 'Add Pipe Node ...'. A table titled 'Select Location in table then select Boundary Condition Type' shows 'River' and 'Reach' selected. Another table, 'Storage/2D Flow Areas', lists 'Perimeter 1' with 'BCLine: Outflow' and 'Normal Depth' boundary conditions. To the right, a 'Normal Depth Downstream Boundary' dialog is open, showing 'Friction Slope' as 0.11 and the option 'Compute separate water surface elevation per face along BC Line' selected.

Boundary Condition Types	
Stage Hydrograph	Flow Hydrograph
Normal Depth	Stage/Flow Hydr.
Lateral Inflow Hydr.	Uniform Lateral Inflow
T.S. Gate Openings	Groundwater Interflow
Elev Controlled Gates	Navigation Dams
Rules	IB Stage/Flow
Precipitation	

Add Boundary Condition Location	
Add RS ...	Add SA/2D Flow Area ...
Add Conn ...	Add Pump Sta ...
Add Pipe Node ...	

Select Location in table then select Boundary Condition Type	
River	Reach
RS	Boundary Condition

Storage/2D Flow Areas	
Perimeter 1	BCLine: Outflow
Perimeter 1	BCLine: Inflow
	Normal Depth
	Flow Hydrograph

Normal Depth Downstream Boundary	
2D: Perimeter 1 BCLine: Outflow	
Friction Slope:	0.11
☐ 2D Flow Area Boundary Condition Parameters ☒ Compute separate water surface elevation per face along BC Line ☐ Compute single water surface for entire BC Line	
OK	Cancel

Figure 27: Outflow BC configuration

Save the unsteady flow data as Unsteady1 and close. After check that the data is imported into the project.

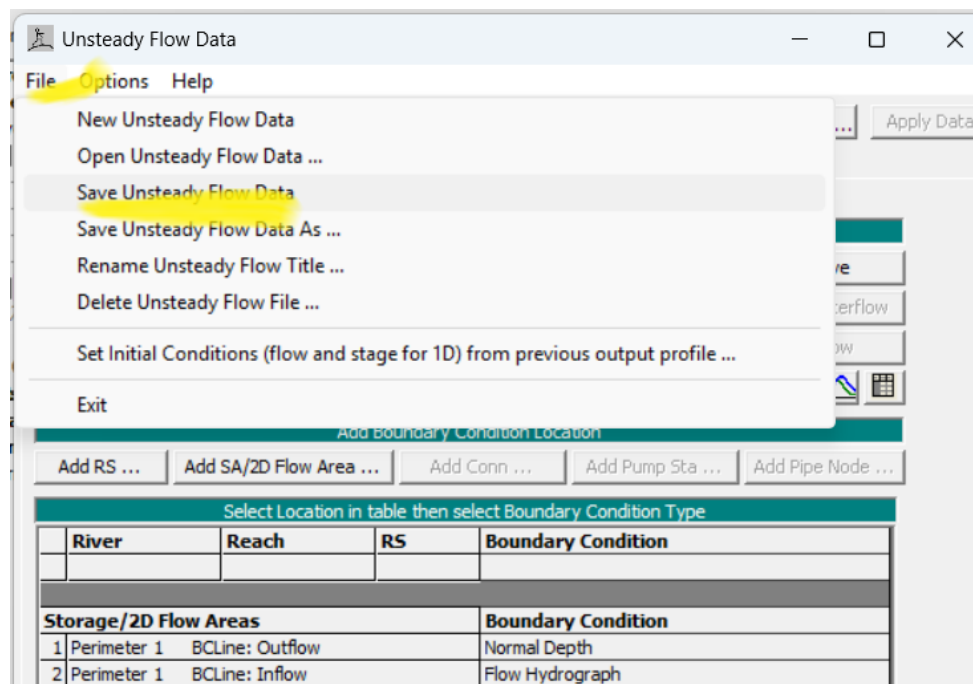


Figure 28: Save the unsteady flow data

5.2 Unsteady Flow Analysis

Now its time to create a plan that compiles the geometry and flow. On Unsteady Flow Analysis fill the following. For the Computation Interval we have to take into account Courant. The cross section area is around 40 m² and flow is around 16m³, so the average velocity is 0.4m/s. If the cell size is 2 meters that means that a particle of water will take 5 seconds to cross it. Therefore the computation interval has to be smaller, we will set up to 4 seconds, otherwise we are going to have unconvergences and wrong simulations.

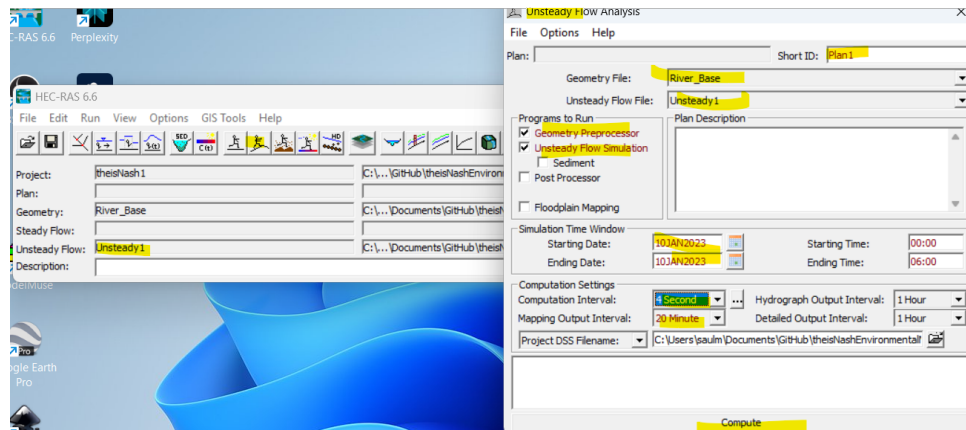


Figure 29: Unsteady flow analysis

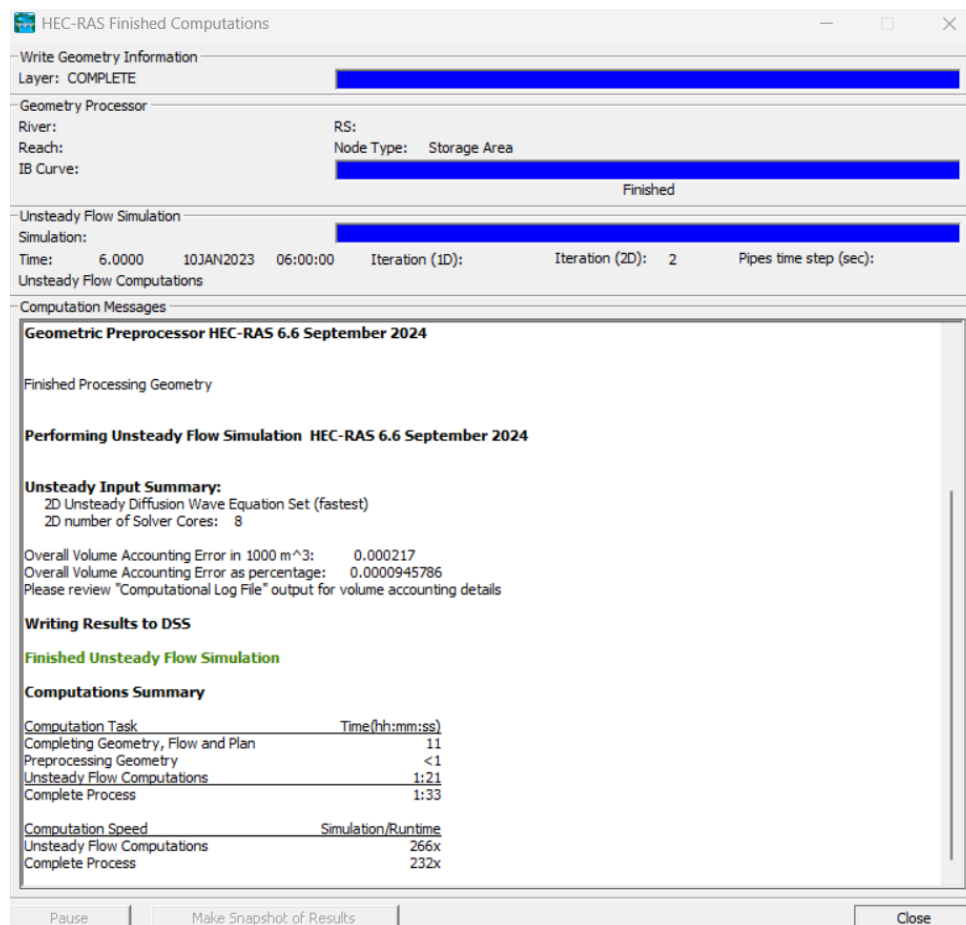


Figure 30: End of the unsteady flow simulations

5.3 Result analysis

Save the plan and open RAS Mapper. Results are already available.

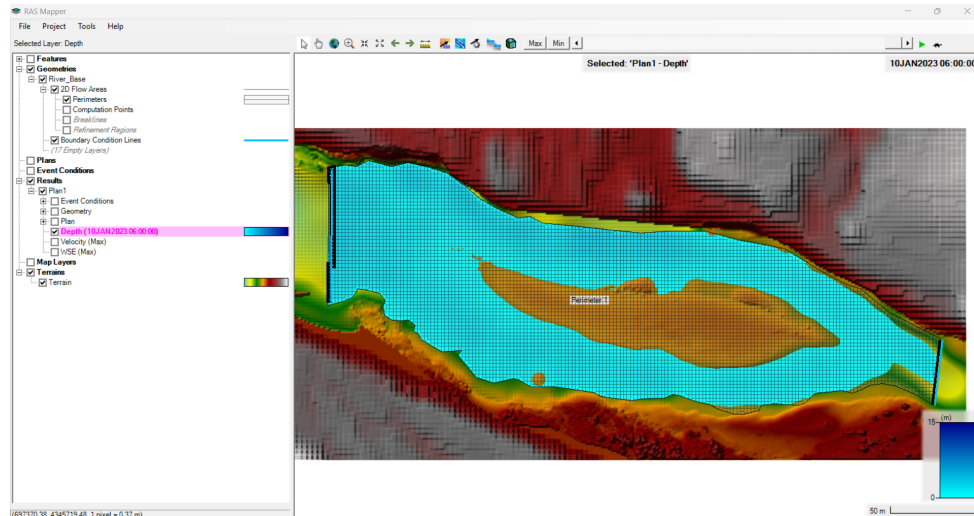


Figure 31: Depth visualization

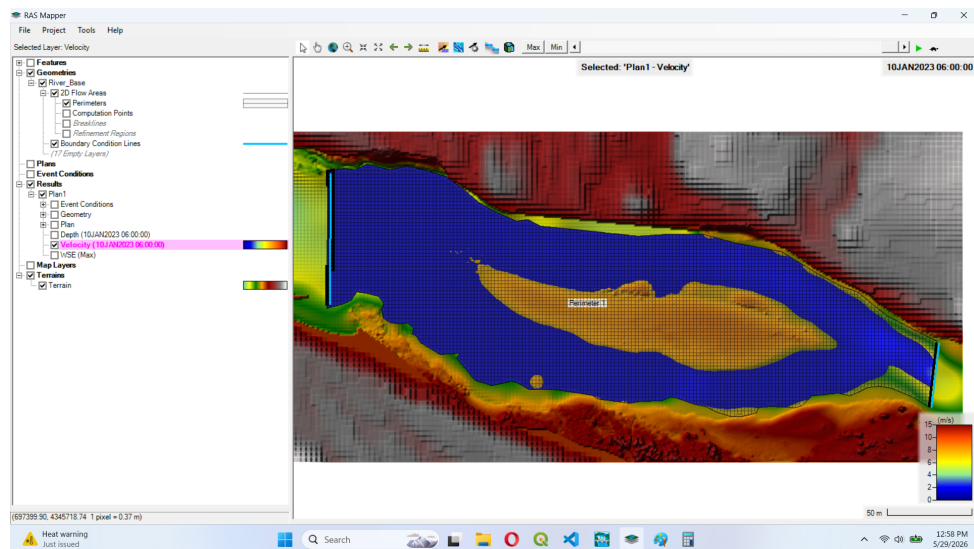


Figure 32: Velocity visualization

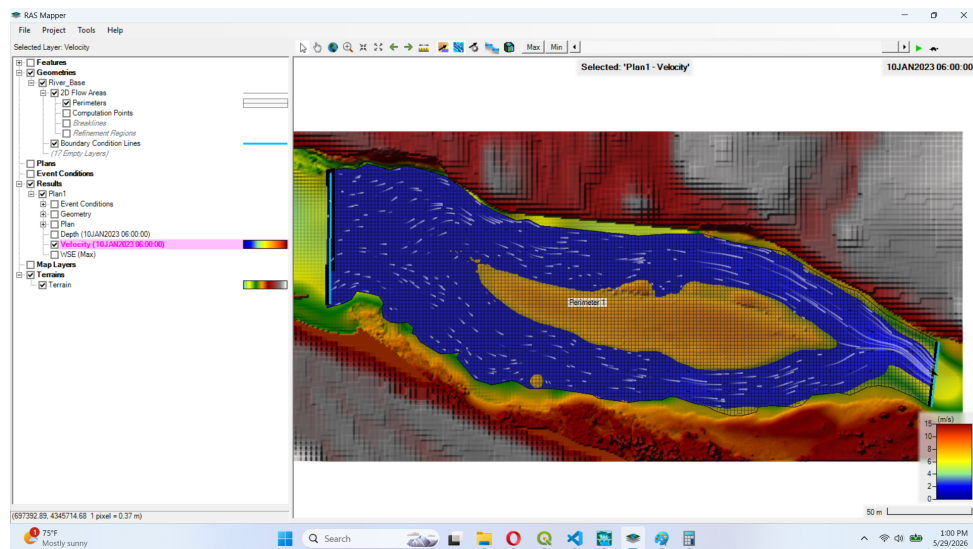


Figure 33: Particle tracking